

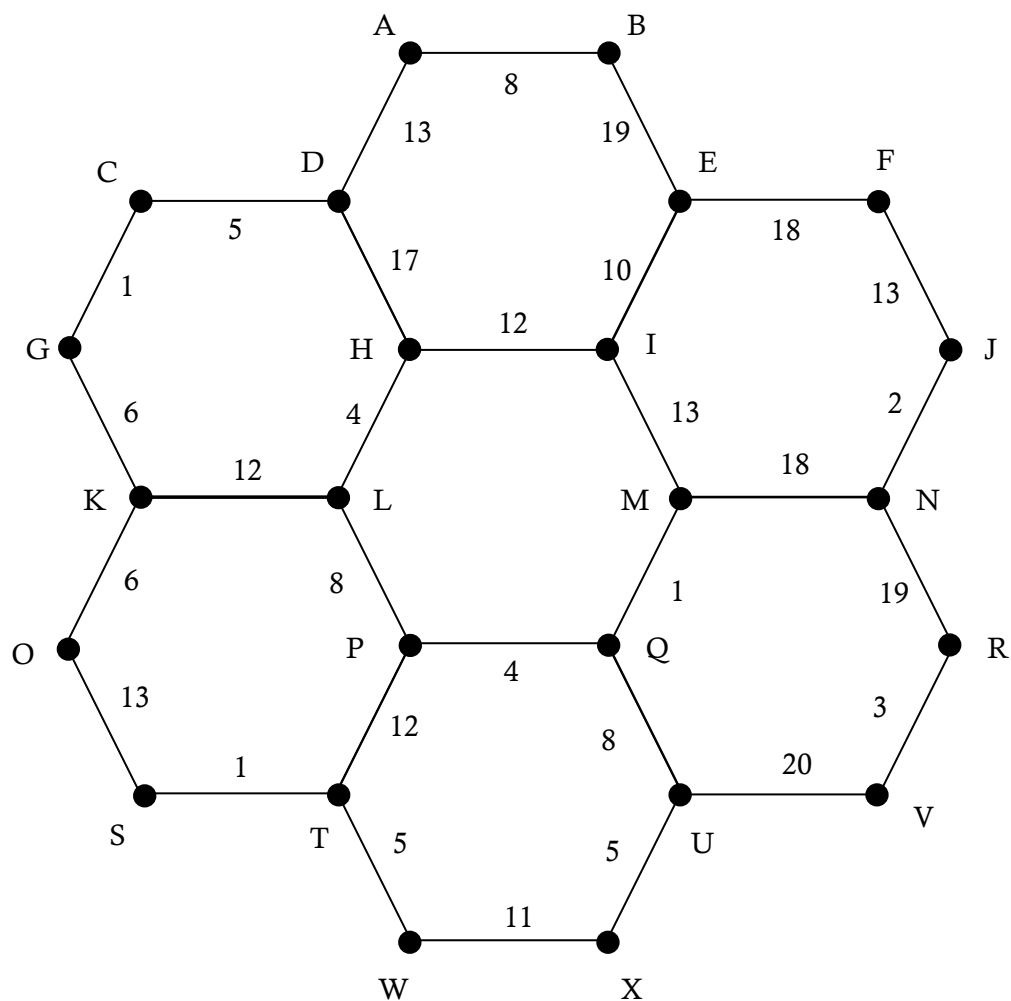
FINDING A PATH OF LEAST COST

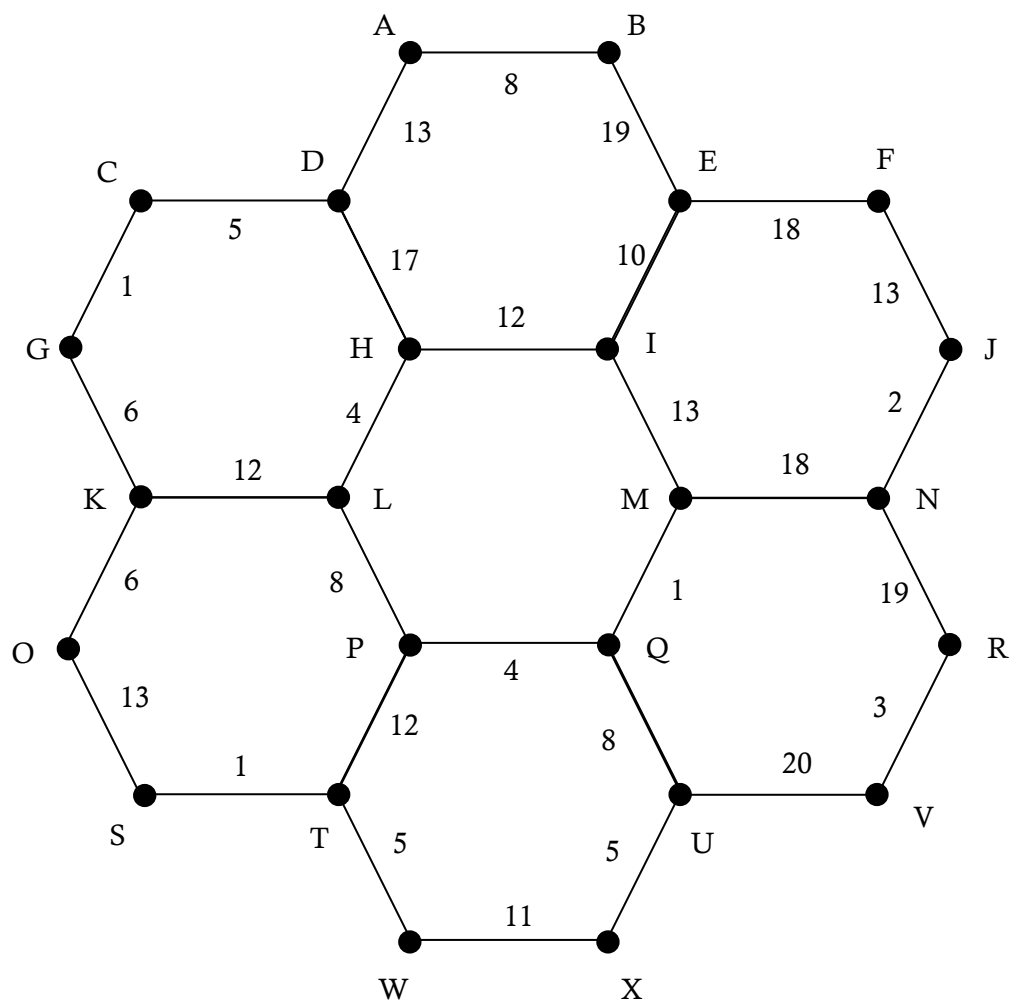
NJS

The network below shows the cost of moving from one vertex to another over different edges.

Investigate algorithms to find the least cost path from vertex A to vertex X. If you have an algorithm that you think works, then use it to find the shortest path between some other pairs of vertices.

There is another copy of the network on the back of this sheet.





Dijkstra's algorithm

START with a graph G . At each vertex draw a box, the lower area for temporary labels, the upper left hand area for the order of becoming permanent and the upper right hand area for the permanent label.

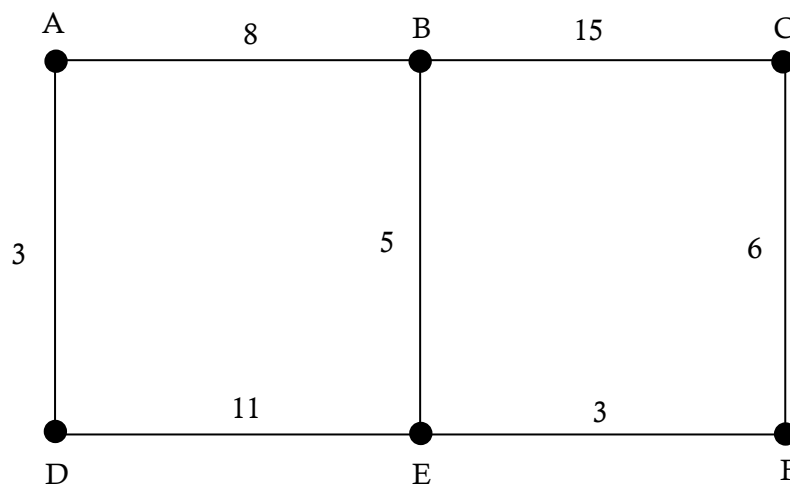
STEP 1 Make the given start vertex permanent by giving it permanent label 0 and order label 1.

STEP 2 For each vertex that is not permanent and is connected by an arc to the vertex that has just been made permanent (with permanent label = P), add the arc weight to P . If this is smaller than the best temporary label at the vertex, write this value as the new best temporary label.

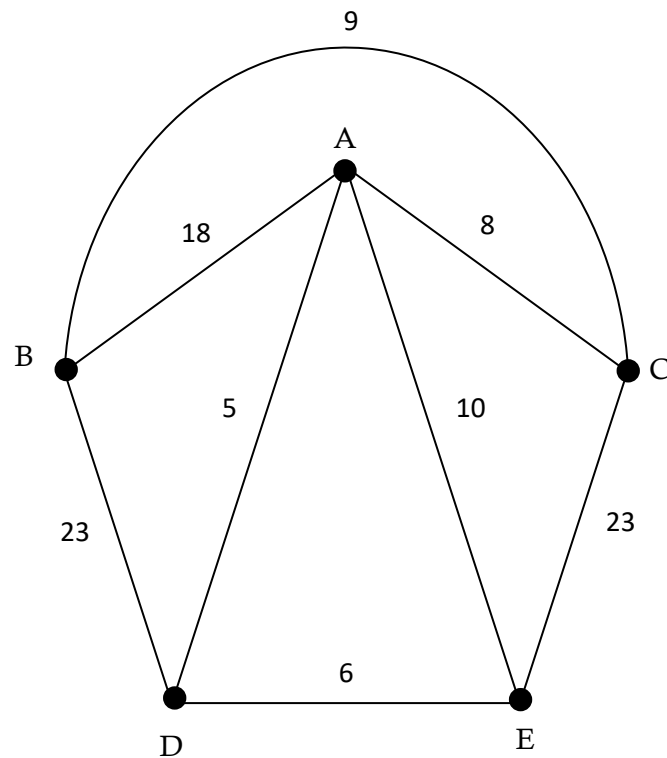
STEP 3 Choose the vertex that is not yet permanent which has the smallest best temporary label. If there is more than one such vertex, choose any one of them. Make this vertex permanent and assign it the next order label.

STEP 4 If every vertex is now permanent, or if the target vertex is permanent, use 'trace back' to find the routes or route, then STOP; otherwise return to STEP 2.

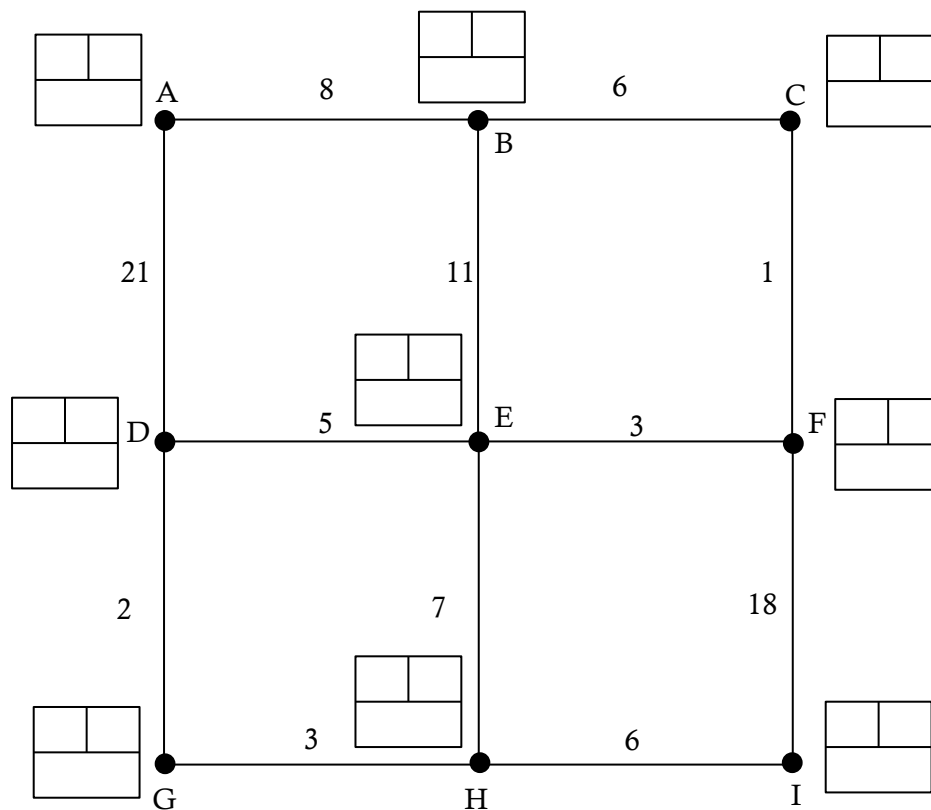
1. Try to apply Dijkstra's algorithm, as given above, to find the lowest cost path from A to F in the network shown below:



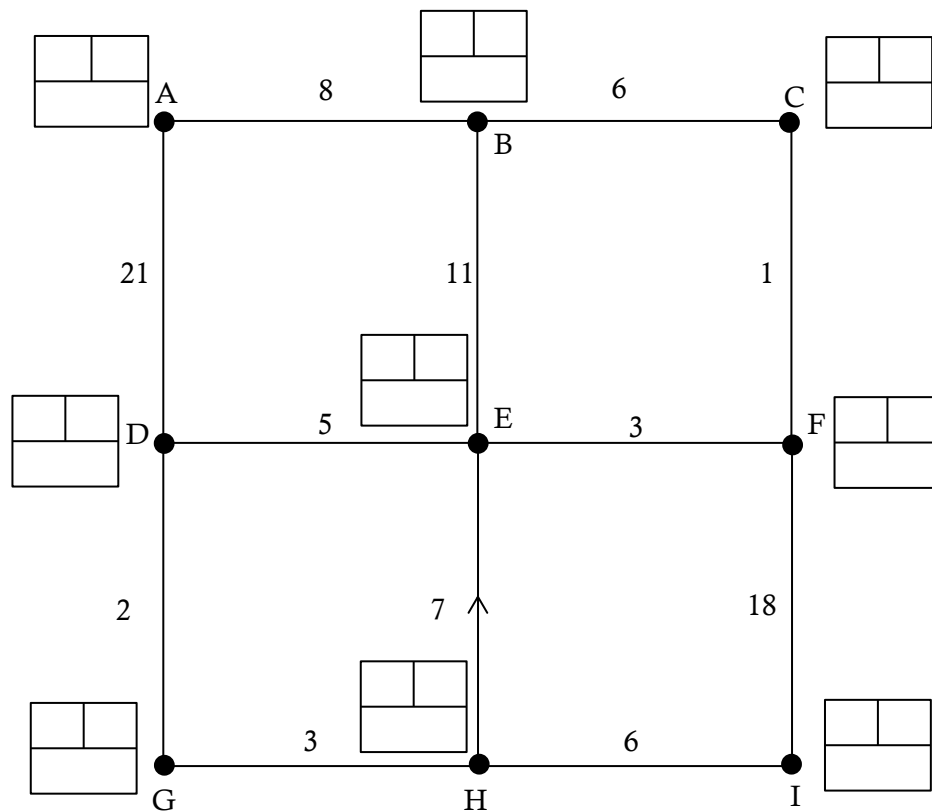
2. [Working as a class] Copy down the steps of Dijkstra's algorithm to find a path of least cost from B to E in the network below:



3. [For discussion] Come up with an informal procedure that establishes the actual lowest cost path once all the boxes, etc., for Dijkstra's algorithm have been completed.
4. Apply Dijkstra's algorithm to find the lowest cost path from A to I in the network below:

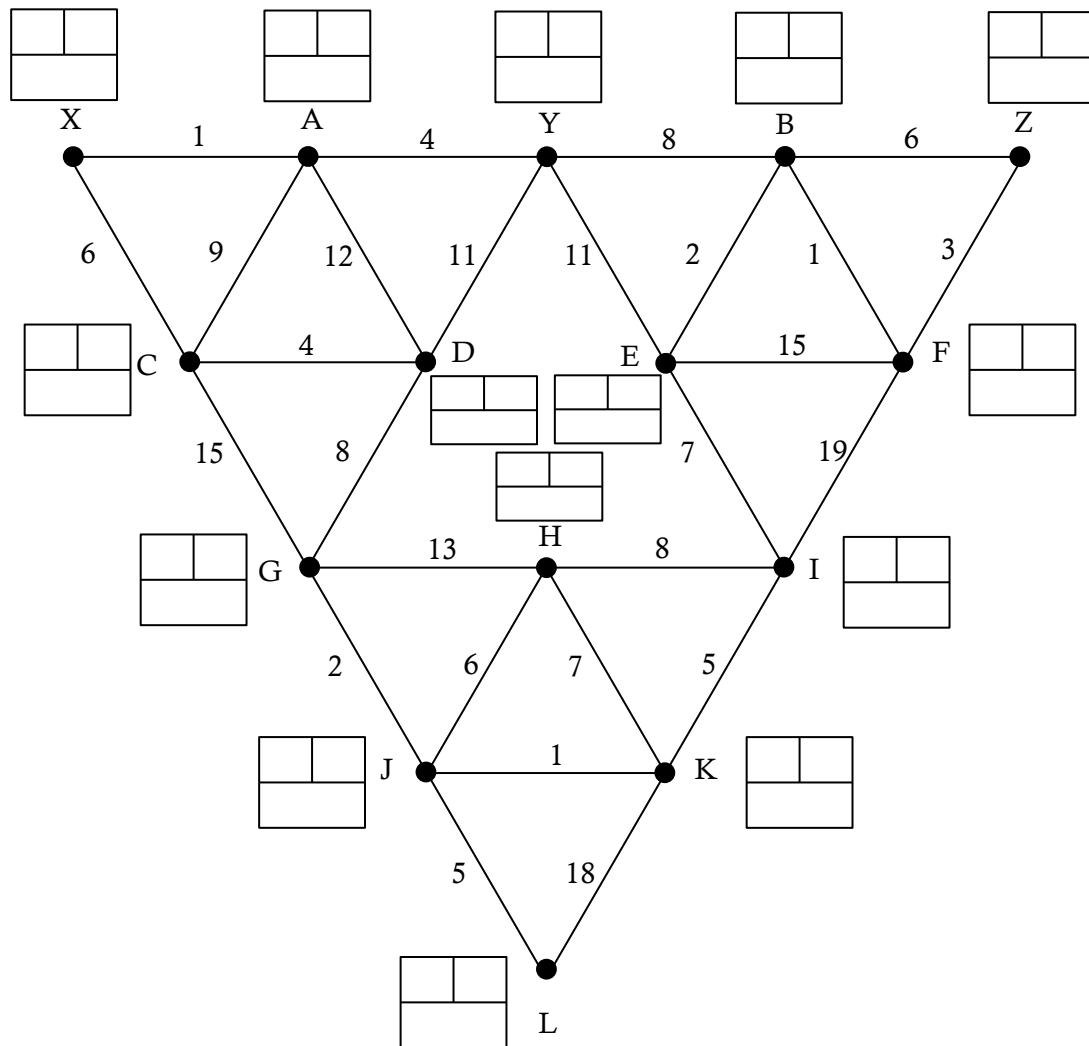


5. In the previous question, suppose that the edge from H to E becomes directed (one-way), as shown below. How will this affect the shortest path?



6. [Reflect] What parts of the output of Dijkstra's algorithm are the same in Questions 4 and 5? What is different?
7. [For discussion] Someone is using Dijkstra's algorithm to find the shortest path from vertex A to vertex Z in a network. At some point, vertex D is made permanent, with an order label of 5. What has been achieved at this stage?

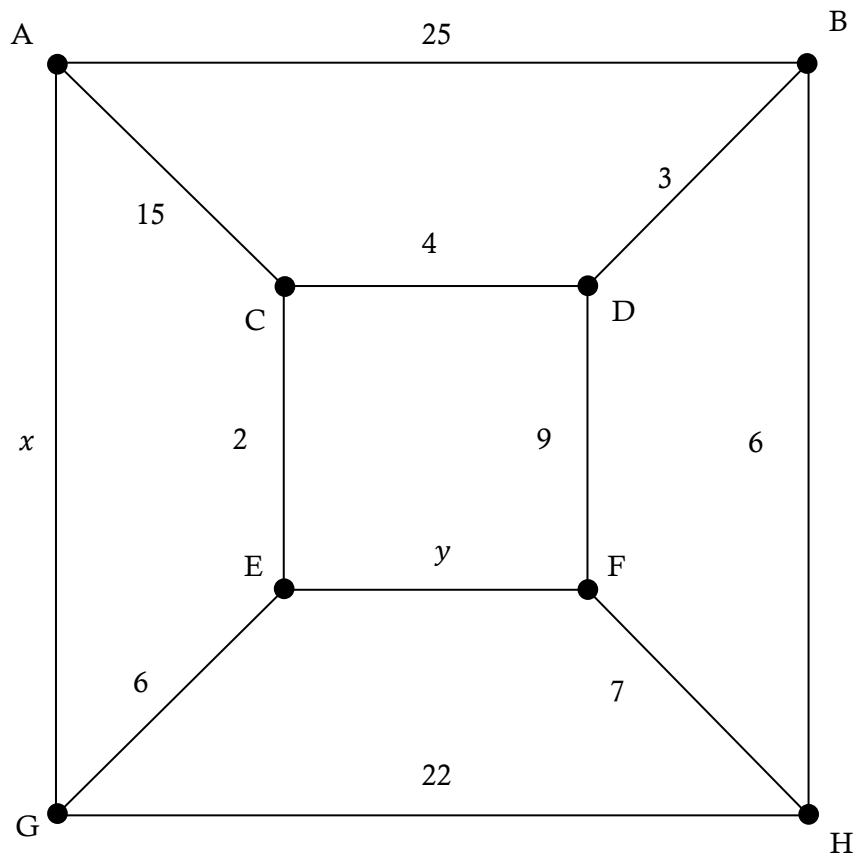
8. The (stylised) road network of Sierpinski County is shown overleaf. Each vertex represents a town, and the values on the arcs are the times required to travel between the towns. There is a fire in town L, but the fire stations are in towns X, Y, and Z. Which town should send it's fire service, in order to arrive at L as soon as possible?



9. Find a counterexample to show that Dijkstra's algorithm might not find a path of least cost if some of the costs are negative.
10. Mark says "If some of the costs in a network are negative then we can add a fixed amount to all of the edges to make all the costs non-negative. We can then apply Dijkstra's algorithm to this new network to find the path of least cost".

Show, by means of a counterexample, that Mark is incorrect.

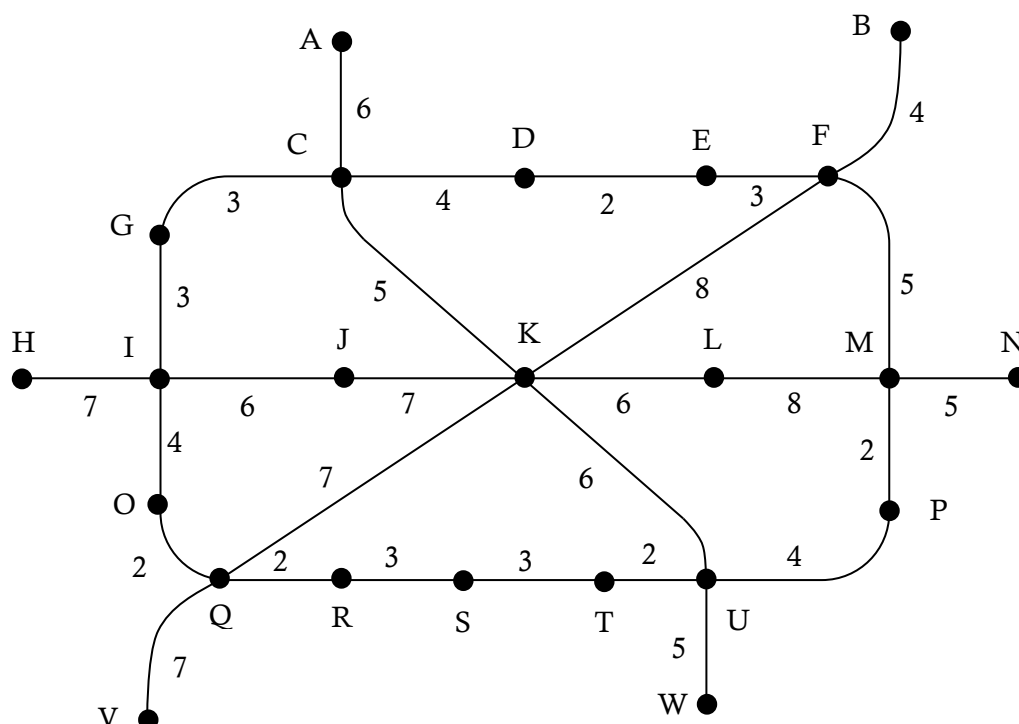
11. In the network below, the path of least cost from A to H needs to be found. Determine the ranges of values of x and y such that edges AG or EF (or both) are included in a least cost path.



12. The graph below shows the underground network for a city. The vertices represent stations, and the weights on the arcs are the average times between adjacent stations.

Apply Dijkstra's algorithm to find the quickest route, on average, from W to H.

[You will need to reproduce a version of this network on file paper. Do not try to crowd an implementation of Dijkstra onto the diagram as it is shown here.]



13. Following on from Question 12, the model is now improved to take account of the time that a train spends at a station letting passengers off and on. Suppose that, on average, a train spends a minute and a half at each station before setting off.

Find a way to adapt the network to take account of this new model and apply Dijkstra's algorithm to find the quickest route, on average, from W to H. Show your working clearly.

14. Following on from Question 13, the model is improved once again to take account of the fact that there are several different underground lines, and that changing trains takes time. There are four lines shown in the above diagram: The rectangle Line, that runs through CFMUQI, the Middle Line, that runs through HIKMN, the Vandyke line, that runs through BFKQV, and the Anniversary Line, that runs through ACKUW.

Suppose that, on average, it takes 6 minutes to change from one line to another (this includes the waiting time for the new train to arrive). That is: in this model, passing through a station on the same line incurs an average delay of 1.5 minutes, and going through a station using different lines incurs an average delay of 6 minutes.

Find a way to adapt the original network diagram of Question 12 to take account of this new model, and apply Dijkstra's algorithm to find the new quickest route, on average.

15. Write a formal algorithm, in flow chart form, that will trace the path of least cost, given a network in which Dijkstra's algorithm has been completed (that is, all the boxes are filled in). Your algorithm should output a list of vertices (describing the shortest path), and use variables as appropriate to keep track of any necessary values.